

3.1.6 Response to Environmental Laws and Regulations

Legal Compliance and Reports on Complaints in FY2021

In fiscal 2021, there was one violation of environment-related laws. The legal violation concerned management of exhaust gas in Taiwan—the preventive measure has already been completed.

	FY2018	FY2019	FY2020	FY2021
Number of legal violations	1	0	1	1
Penalty (USD)	4,000	0	168	3,600
No. of cases addressed	1	0	1	1
Penalty paid (USD)	4,000	0	168	3,600
No. of cases being addressed	0	0	0	0
Debts as of the end of fiscal year (USD)	0	0	0	0
No. of other violations and complaints	2	2	0	0

Responses to Environment-Related Complaints and Legal Violations in FY2022*

Company/Site Name	Description	Responses
FUJIFILM Electronic Materials Taiwan Co., Ltd.	Violation of Air Pollution Control Act	Enhancing exhaust gas cleaning facility and improving monitoring frequency

* Relatively minor violations have been excluded.

3.1.7 Environmental Education for Employees

To promote our group-wide environmental activities, the Fujifilm Group provides environmental education for all employees to acquire basic environmental knowledge and raise awareness for environmental issues. The education program is provided continuously through the e-learning system to encourage understanding and implementation of the Fujifilm Group Green Policy and SVP2030.

Although the contents of such education programs previously varied from country to country, we decided to take a united approach across the Group from fiscal 2021 to further encourage our group-wide environmental activities. We firstly conducted the same presentation and surveys under the theme of “Addressing climate change” for Fujifilm Group employees in Japan and China and plan to expand the coverage of the same program to Europe and other regions in 2022.

For employees who require specific environmental knowledge and skills, we also offer specialized education programs, including those about management of chemical substances, Design for the Environment, waste management, and product safety.

Environmental Education  <https://holdings.fujifilm.com/en/sustainability/activity/environment/education>

3.1.8 Environmental Management in the Supply Chain

The Fujifilm Group considers the impact that our business has on the environment throughout its lifecycle, from the procurement of raw materials to the manufacturing, transportation, use, and disposal of products, and is working throughout our supply chain to reduce the environmental impact. For further information on the Group's basic policy and specific cases, please refer to 2.5.6 *Environmental Efforts in the Supply Chain*.

3.2 Climate Change Strategy

3.2.1 Basic Approach

In December 2021, the Fujifilm Group set new CO₂ emissions reduction targets to build a decarbonized society. These new targets mean that we will achieve zero CO₂ emissions from our energy consumption*¹ by fiscal 2040 by maximizing the efficiency of our energy use and by using renewable energy sources. We will also reduce CO₂ emissions across the entire product lifecycle—from the procurement of raw materials to the manufacture, transportation, use and disposal—by 50% (compared to fiscal 2019 levels) by fiscal 2030. The new decarbonization targets established by the Fujifilm Group comply with the 1.5°C goal set out in the Paris Agreement.

To achieve these targets, we have established a new Fujifilm Group environmental strategy: “Green Value Climate Strategy.”*² Under the strategy we promote manufacturing of lower environmental burden by introducing and utilizing fuels that do not emit CO₂ substantially such as synthetic methane and hydrogen in addition to electric power. Also we promote development and dissemination of products and services with excellent environmental performance. Additionally, we have started internal carbon pricing (ICP) system to accelerate the implementation of these measures in fiscal 2022. We continue to direct our strategy and efforts to initiatives that address the imminent global social issue of climate change.

*¹ Direct emissions from our manufacturing processes (Scope 1) and indirect emissions due to the use of electricity and steam supplied from other companies (Scope 2).

*² Please refer to the following site for Green Climate Strategy

Environmental Strategy Briefing on April 13, 2022

 https://ir.fujifilm.com/en/investors/ir-materials/presentations/main/0118/teaserItems1/0/tableContents/019/multiFileUpload2_0/link/ff_presentation_20220413_001j.pdf

3.2.2 Governance

Our activities related to climate change are deliberated on and determined by the ESG Committee, chaired by the president. The Committee is held on a regular basis, and the agenda is then reported to the Board of Directors, who then issue instructions and advice in response to the report received from the ESG Committee to ensure the effectiveness of the process.

Issues related to climate change are deliberated on by the ESG Committee as priority risks along with other issues on compliance and other risks (2.2.3 *Compliance and Risk Management Promotion Structure*). In addition to CO₂ emissions reduction and renewable energy introduction targets, the Committee has decided in the past to participate in climate change-related initiatives, including endorsing the TCFD recommendations, joining RE100 and obtaining SBT validation.

In fiscal 2021, the introduction of internal carbon pricing (ICP) and information disclosure in compliance with the TCFD recommendations were deliberated on by the Board of Directors. Additionally, the effect that the degree of achievement of decarbonization targets has on medium-term performance-linked share-based remuneration is determined through deliberations by the ESG Committee and the final decision is taken by the Board of Directors.

3.2.3 Risk Management

The Fujifilm Group has in place an IT system to monitor all group companies' environmental performance related to climate change. The system enables us to collect emission volumes of greenhouse gases such as CO₂ and CFCs and energy consumption volumes at each of our business sites in various countries and regions and use the data to identify risks. The Energy Strategy Promotion Committee analyzes factors affecting these risks and reports the high-priority issues to the ESG Committee, and consequently supports them in deciding on the appropriate operations. We have utilized internal carbon pricing to evaluate climate-related risks, and have reviewed the anticipated impacts and action required.

A scenario analysis based on the TCFD recommendations enables us to identify the risks derived from our environmental performance as well as the risks happening in our supply chain and site locations. We take measures to deal with important issues to be implemented.

3.2.4 Strategy

For TCFD scenario analysis, we have taken consideration into two types of climate-related scenarios in reference to the Representative Concentration Pathways (RCP) 2.6 through 8.5 described in the report of the Intergovernmental Panel on Climate Change (IPCC). One is the 1.5°C scenario in which rigorous measures are implemented to build a decarbonized society and effectively keep temperature rises down to 2°C or less than the level during the Industrial Revolution by the year

2100. The other is the 4°C scenario in which the average temperature predicted to rise by 3.2 to 5°C over the level during Industrial Revolution as a result of failing to implement measures that surpass the current ones.

1. Scenario analysis results

(1) 4°C scenario:

Measures that surpass the current ones have not been implemented, resulting in the average temperature predicted to rise by 3.2 to 5°C over the level during the Industrial Revolution by the year 2100.

① Risks (Physical risks)

In the 4°C scenario, risks such as the impact of climate change on production facilities, disruption of raw materials supply, factory shutdowns due to power failures, etc., were identified. To address these risks, we have been developing preventive measures based on our Business Continuity Plans (BCPs), which include diversification of production sites and procurement sources and implementation of measures to secure stable power supplies. Especially in recent years, typhoons and heavy rains caused by climate change have caused damage by disrupting power grids, which form an important lifeline, in various regions. Since the 1960s, the Fujifilm Group has been avoiding the risk of operational shutdowns due to power failures by sequentially setting up in-house cogeneration systems at its major production sites, thereby securing stable power supplies.

The risk of changes to animal and plant habitats caused by temperature increases and changes in precipitation patterns may lead to a decline in the animal and plant population and to extinction. These changes are likely to cause procurement instabilities and steep rises in prices for plant-derived materials. In addition, depletion of fossil fuel resources is expected to cause supply instabilities and rising prices for petroleum-based materials. The Fujifilm Group is working to reduce these risks by reducing its material consumption, including employing thinner films made from plant-derived materials and recycling parts of multifunction devices in the area of business innovation.

② Opportunities

Global warming has increased the frequency and severity of extreme heat events, marine heat waves, rainstorms, droughts and tropical cyclones. We anticipate a rise in the social demand for products and services aimed at adapting to extreme weather patterns and the impact on ecosystems and human health associated with such extreme weather.

[Resilience of social infrastructure]

In the face of the growing frequency of such extreme weather, resilience of the social infrastructure is one of the important issues that require action. To adapt to these climate changes, the Fujifilm Group is committed to contributing to society with its high-definition lens processing and manufacturing technologies to create high-sensitivity surveillance cameras for monitoring river and sea levels at night and during inclement weather and with its degradation analysis technologies for bridges and embankments that makes use of high-precision image analysis and AI technologies. In addition, we foresee a rise in the importance of solutions that contribute to early disaster relief and restoration of the livelihood of citizens in the event of a disaster, as well as the maintenance of local government functions, through digitization of disaster response processes for local governments.

[Securing drinking water and water for agricultural use]

Changes in the climate and ecosystem will also cause difficulties in accessing drinking water and food. This is expected to result in an increase in drinking water production and manufacturing at plant factories. The global water shortage is growing into a major risk, and we believe we can make a significant contribution to securing drinking water and water for agricultural use with our filtration technology, which employs ion exchange membranes for the desalination of brine and seawater.

[Reducing burdens on medical professionals and improving accessibilities to medical services]

The rise in temperature is having a serious impact on people's health. The burden on medical professionals has increased with the unforeseen spread of infectious and various other diseases. Typhoons, torrential rains and heat waves have increased in frequency, impeding the movement of patients and medical professionals. The possibility of the collapse of the medical care system is a real risk in countries and regions suffering from a shortage of medical professionals. Applying our medical IT technologies and diagnostic imaging systems and AI technologies on a global scale will enable us to contribute to reducing the burden on medical professionals and to improve medical accessibility through remote diagnosis and other methods.

(2) 1.5°C scenario

Rigorous measures are implemented to build a decarbonized society, keeping temperature rises down to 2°C or less than the level during the Industrial Revolution by the year 2100.

① Risks (Physical risks)

In the 1.5°C scenario, measures to cut the use of fossil fuels and foster technological innovation are implemented in the transition to a decarbonized society. They include, for example, a carbon tax and carbon border adjustments mechanism designed to discourage the transfer of business operations prompted by differences in carbon taxes among countries and

(3) Task Force for Climate-related Financial Disclosure (TCFD)

An organization that advises corporations on disclosing financial information concerning climate change risks and opportunities. We announced our endorsement of the TCFD recommendations in December 2018.

<Our activities>

- Joined the TCFD Support Project by the Ministry of the Environment in 2019.
- Joined the TCFD Consortium in 2019.
- Began disclosing environmental analyses and information based on TCFD recommendations starting with our Sustainability Report 2019.

(4) RE100

An international initiative of corporations that are committed to using 100% renewable electricity for their business operations. We joined the organization in 2019.

<Participation background: Decarbonization difficulties in the chemical industry>

The chemical industry requires high temperature steam heat. It is extremely difficult to replace the conventional energy required for these processes with renewable energy using current technology. We are aware that acceleration of technological innovation is indispensable towards the decarbonization of electricity and energy, not only within our companies but across society as a whole. For this reason, we joined RE100 in 2019, aiming at zero CO₂ emissions by 2050 through replacement with fuels without CO₂ emissions, such as hydrogen, in addition to our renewable energy usage target. In 2021, we increased the target values and brought forward the goal date to 2040.

<Our activities>

- In June 2019 at the JCLP Symposium, the “Proposal from Corporate Consumers Aiming to Source 100% of Power from Renewable Electricity” was released based on discussions among RE100 Japanese companies.
- In 2020, in a series of seminars, “Corporate Sourcing of Renewable Energy,” jointly held by RE100 and GWEC/CDP, we presented the Fujifilm Group’s practice of renewable energy procurement.
- In March 2021, under a collaboration with JCLP, 53 of the RE100 companies submitted a letter calling for the Japanese government to increase their renewables ambitions.

(5) Science-Based Targets initiative (SBTi)

A partnership comprising the World Wide Fund for Nature (WWF), CDP (former Carbon Disclosure Project), United Nations Global Compact, and the World Resources Institute (WRI). The initiative recommends corporations to set science-based CO₂ emissions reduction targets to limit the global average temperature rise to 1.5°C above pre-industrial levels.

<Our activities>

- In 2017, we announced our reduction targets and obtained the official 2°C Limit validation within the same month.
- In 2020, we obtained the official Well-Below 2°C Limit validation by presenting our revised targets.

2. Endorsement and support for climate change policies

The Fujifilm Group announces political requests and endorsements through participating in climate change initiatives and industry organizations.

<Our activities>

- 2019: Participated in the “Project to Support the Use of Scenario Analysis in line with TCFD recommendations” by the Ministry of the Environment.
- 2020: Participated in the Ministry of the Environment’s project to promote a decarbonized society, “Spreading Carbon Neutral – Senior Management Talks about Decarbonization” through our video. The video introduced the Fujifilm Group’s anti-climate change efforts, such as the 100% switch to wind energy in FUJIFILM Manufacturing Europe B.V. in the Netherlands in 2016, and setting group targets for renewable energy usage in 2019.
- Participating in projects by the New Energy and Industrial Technology Development Organization (NEDO), “Development of Continuous Production and Process Technologies of Fine Chemicals” and “Artificial Photosynthesis.” We were listed among the corporations under the Zero Emissions Challenge 2021, which covers companies participating in projects related to the Environment Innovation Strategy, under the Ministry of Economy, Trade and Industry.
- Keidanren Carbon Neutrality Action Plan (former Commitment to a Low Carbon Society)
Cross-border voluntary efforts by the business community aiming at carbon neutrality by 2050. The plan is designed to encourage corporations to create their carbon neutrality visions and monitor their progress by undertaking checking and assessment. We participated in the plan through the Japan Chemical Industry Association by annually reporting progress in CO₂ emissions reduction and energy-saving measures.

<Our concrete measures>

- We are working to reduce CO₂ emissions by continuing our pursuit of efficient energy usage in production processes, begun in the 1990s, and a gradual switch to low carbon fuels, such as natural gas, since the 2000s.
- The manufacturing industry tends to aim at optimization of production efficiency towards a set CO₂ emissions intensity target. However, we decided to set an absolute reduction target determined from the base year to achieve business expansion and emissions reduction at the same time.

3.3.3 Data Related to Climate Change Measures

1. GHG Emissions (Scope 1, 2)

Annual Changes

	Unit		FY2017	FY2018	FY2019	FY2020	FY2021
Total direct GHG emissions (Scope 1)	kt CO ₂ e	Fujifilm Group Total	687	672	633	603	633
		Fujifilm	644	629	594	567	599
		FUJIFILM Business Innovation	43	43	40	36	34
	Unit		FY2017	FY2018	FY2019	FY2020	FY2021
Total indirect GHG emissions (Scope 2) Market-based	kt CO ₂ e	Fujifilm Group Total	531	510	451	409	420
		Fujifilm	379	366	325	294	306
		FUJIFILM Business Innovation	152	143	126	115	114
	Unit		FY2017	FY2018	FY2019	FY2020	FY2021
Total indirect GHG emissions (Scope 2) Location-based	kt CO ₂ e	Fujifilm Group Total	593	576	514	475	494
		Fujifilm	436	427	382	354	375
		FUJIFILM Business Innovation	157	149	132	120	118

* Scope 1: Total of vehicle and fuel-derived CO₂ emissions plus chlorofluorocarbons.

* Scope 2: The 2021 edition IEA factors were used to calculate location-based emissions. Excludes internal power distribution from private power generators and cogeneration systems.

* The coverage of each data item is for 100% of total sales.

* The above "Scope 1 and 2 (market-based)" data is verified by a third party organization (SGS Japan Inc.).

2. GHG Emissions (Scope 3)

FY2021 Results

No.	Category	Emissions from the Fujifilm Group (kt CO ₂ e)	Proportion of Scope 3	Fujifilm	FUJIFILM Business Innovation
1	Purchased goods and services	2,130	61.3%	1,462	667
2	Capital goods	441	12.7%	348	93
3	Energy excluding Scope 1,2	144	4.1%	136	8
4	Transportation and distribution (Upstream)	12	0.3%	6	6
5	Waste generated in operation	9	0.3%	8	1
6	Business travel	10	0.3%	5	5
7	Employee commuting	29	0.8%	15	14
8	Leased assets (Upstream)	Calculated as Scope 2	—	—	—
9	Transportation and Distribution (Downstream)	286	8.2%	125	161
10	Processing of sold products	77	2.2%	77	—
11	Use of sold products	90	2.6%	90	Calculated as Category 13
12	Disposal of sold products	88	2.5%	88	—
13	Leased assets (Downstream)	158	4.5%	—	158
14	Franchise	N/A	—	—	—
15	Investment	N/A	—	—	—
	Total	3,473	100%	2,360	1,113

3. GHG Emissions across the Entire Product Lifecycle

FY2021 Result of GHG Emission

Unit: kt-CO₂/year

	Procurement		Manufacturing		Transportation	Use		Disposal	Total
	2,273		1,053		286	248		88	3,948
Items	PET, TAC, etc.	458	Fuels	596	286	Medical equipment	62	88	
			Vehicles	25		Minilab	20		
	Aluminum	1,148	Fluorocarbons	12		Copiers/Printers/Fax machines	158		
			Purchased electric power	408					
	Copiers/Printers/Fax machines	667	Purchased heat	12		Others	8		

4. Energy Consumption Data*1

Annual Changes

Unit: TJ

Scope		FY2017	FY2018	FY2019	FY2020	FY2021
Heavy oil, etc.*1	Fujifilm	1,350	1,134	1,049	920	905
	FUJIFILM Business Innovation	6	4	3	2	2
	Group Total	1,356	1,139	1,053	923	907
Vehicles	Fujifilm	130	130	126	116	121
	FUJIFILM Business Innovation	330	301	273	257	252
	Group Total	461	431	400	374	373
Gas*2	Fujifilm	10,427	10,436	9,932	9,654	10,297
	FUJIFILM Business Innovation	355	362	348	304	266
	Group Total	10,782	10,798	10,280	9,959	10,563
Bio Gas*3	Fujifilm	39	55	59	53	54
	FUJIFILM Business Innovation	0	0	0	0	0
	Group Total	39	55	59	53	54
Purchased electric power	Fujifilm	8,263	8,170	7,876	7,152	7,285
	FUJIFILM Business Innovation	2,747	2,606	2,372	2,167	2,140
	Group Total	11,010	10,776	10,248	9,319	9,425
Purchased heat	Fujifilm	72	98	90	83	120
	FUJIFILM Business Innovation	122	114	115	105	114
	Group Total	194	213	205	188	233
Renewable Energy*4	Fujifilm	839	794	770	785	976
	FUJIFILM Business Innovation	0	0	0	0	0
	Group Total	839	794	770	785	976
Fujifilm		21,121	20,818	19,902	18,765	19,756
FUJIFILM Business Innovation		3,561	3,388	3,111	2,836	2,775
Group Total		24,682	24,206	23,014	21,601	22,531

* The actual sum may not match the total of the displayed values due to how the values are handled.

*1 Total of heavy oil A, heavy oil C, kerosene oil, light oil, and gasoline.

*2 Total of natural gas, liquified natural gas, city gas, butane, and liquified petroleum gas.

*3 Landfill methane gas.

*4 Total of power from private generators and purchased energy.

5. Renewable Energy-derived Power Consumption

Annual Changes

Unit: MWh

		FY2017	FY2018	FY2019	FY2020	FY2021
Renewable energy-derived power consumption	Fujifilm Group Total	99,907	94,741	94,658	94,038	112,728
	Fujifilm	99,907	94,741	94,658	94,038	112,728
	FUJIFILM Business Innovation	0	0	0	0	0

* The data items in fiscal 2021 and earlier were also amended because the criteria for inclusion as renewable energy was revised.